concurrently to Process 3.0 (NLP Contextual Analysis) and Process 4.0 (Deep Learning Pattern Detection), which are also calibrated against D1. Independently, Process 5.0 (Anomaly Detection) evaluates behavioural data drawn from the behavioural logs data store (D2). The outputs of Processes 2.0, 3.0, 4.0 and 5.0 are passed to Process 6.0 (Risk Scoring), which computes a combined risk score. This result is passed to Process 7.0 (Explainable AI Reporting) and Process 8.0 (Alert and Dashboard), whose combined output, the risk score, classification, explanation and any alert, is delivered to the security analyst.

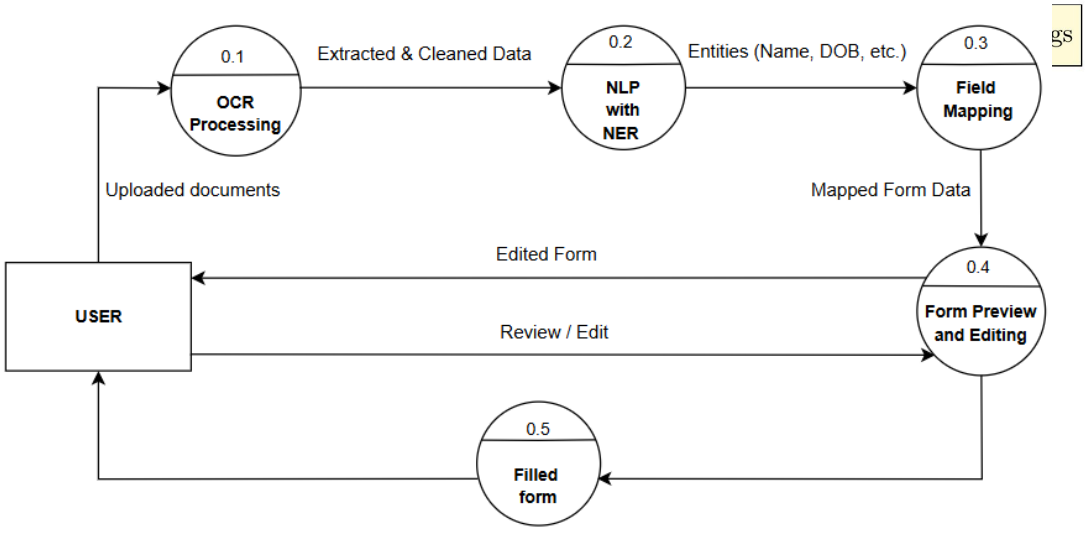


Fig. 5.5: DFD Level 1 of the Intelligent Personal Data Leakage Detection System

### 5.2.3 DFD Level 2

To clarify the internal steps of the two most composite processes identified at Level 1, this section expands Process 2.0 (Rule-Based Filtering) and Process 6.0 (Risk Scoring) into their constituent sub-processes.

Figure 5.4 expands Process 2.0. The validated input received from Process 1.0 is passed to Sub-process 2.1 (Load Regex Rule Set), which retrieves the configured pattern definitions from the rule configuration store (D1). Sub-process 2.2 (Pattern Matching Engine) then scans the input against the loaded rule set, and Sub-process 2.3 (Match Tagging and Confidence Scoring) tags each match with its pattern type and a confidence value before the detected patterns are forwarded to Process 6.0.